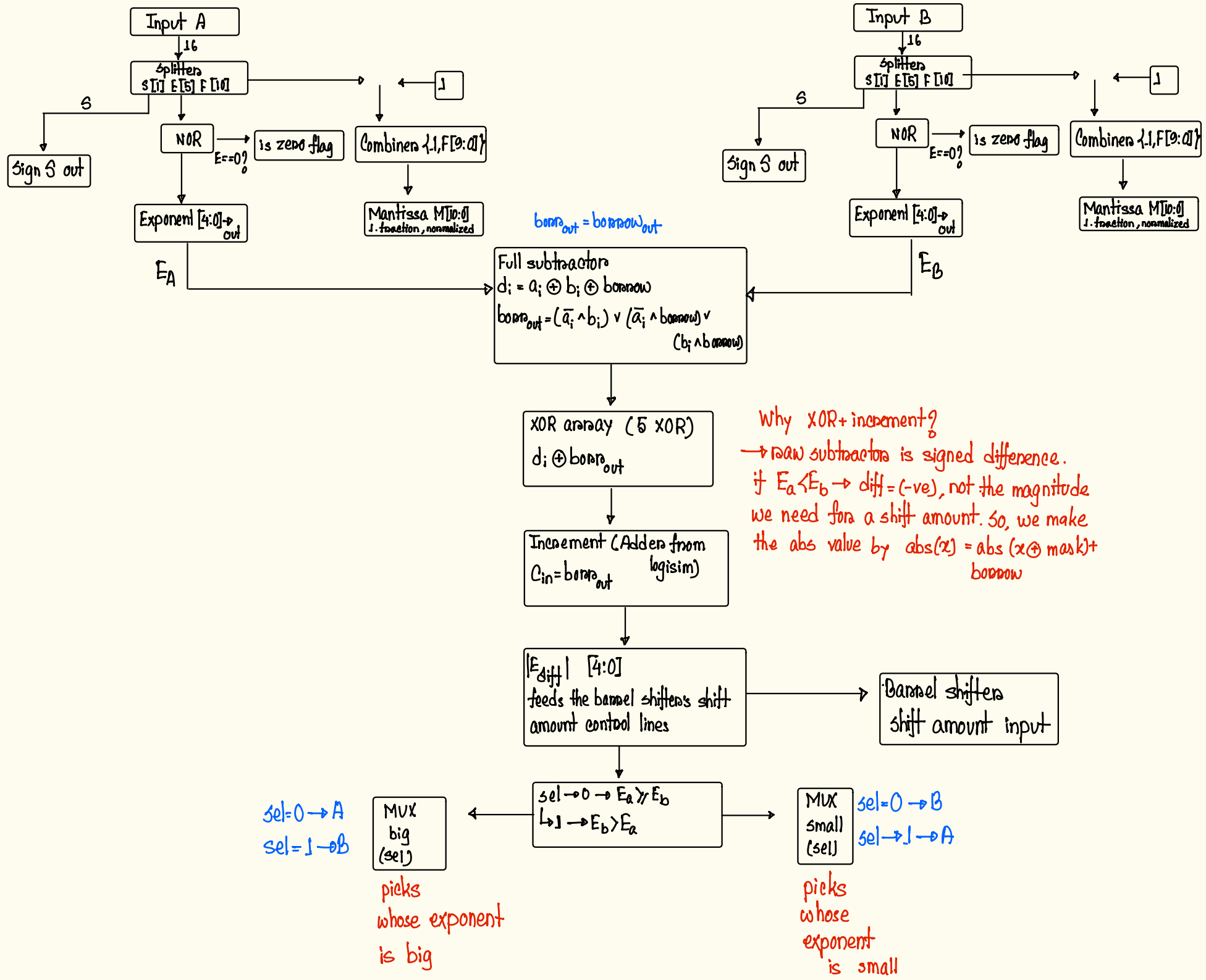
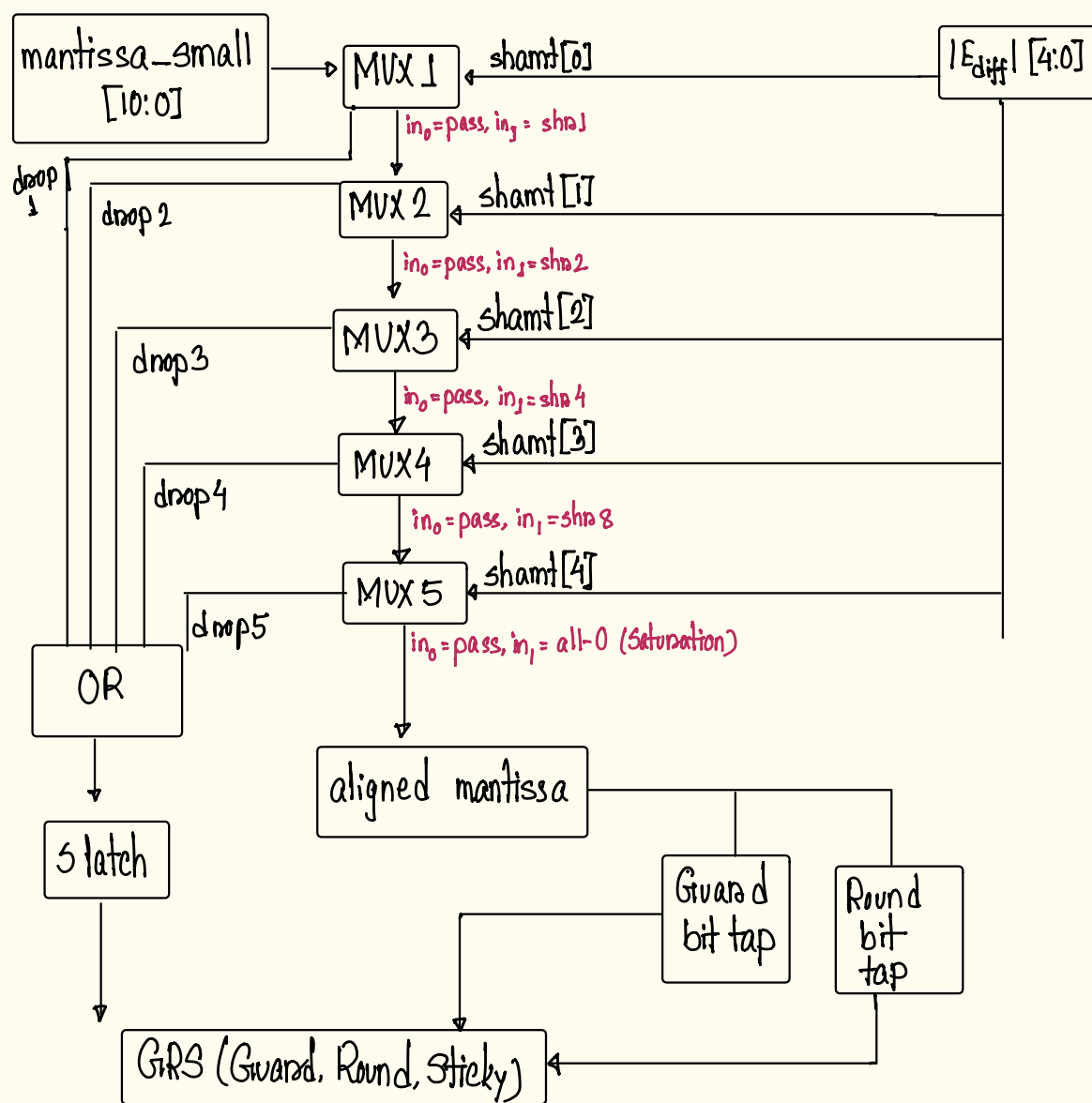






MUX

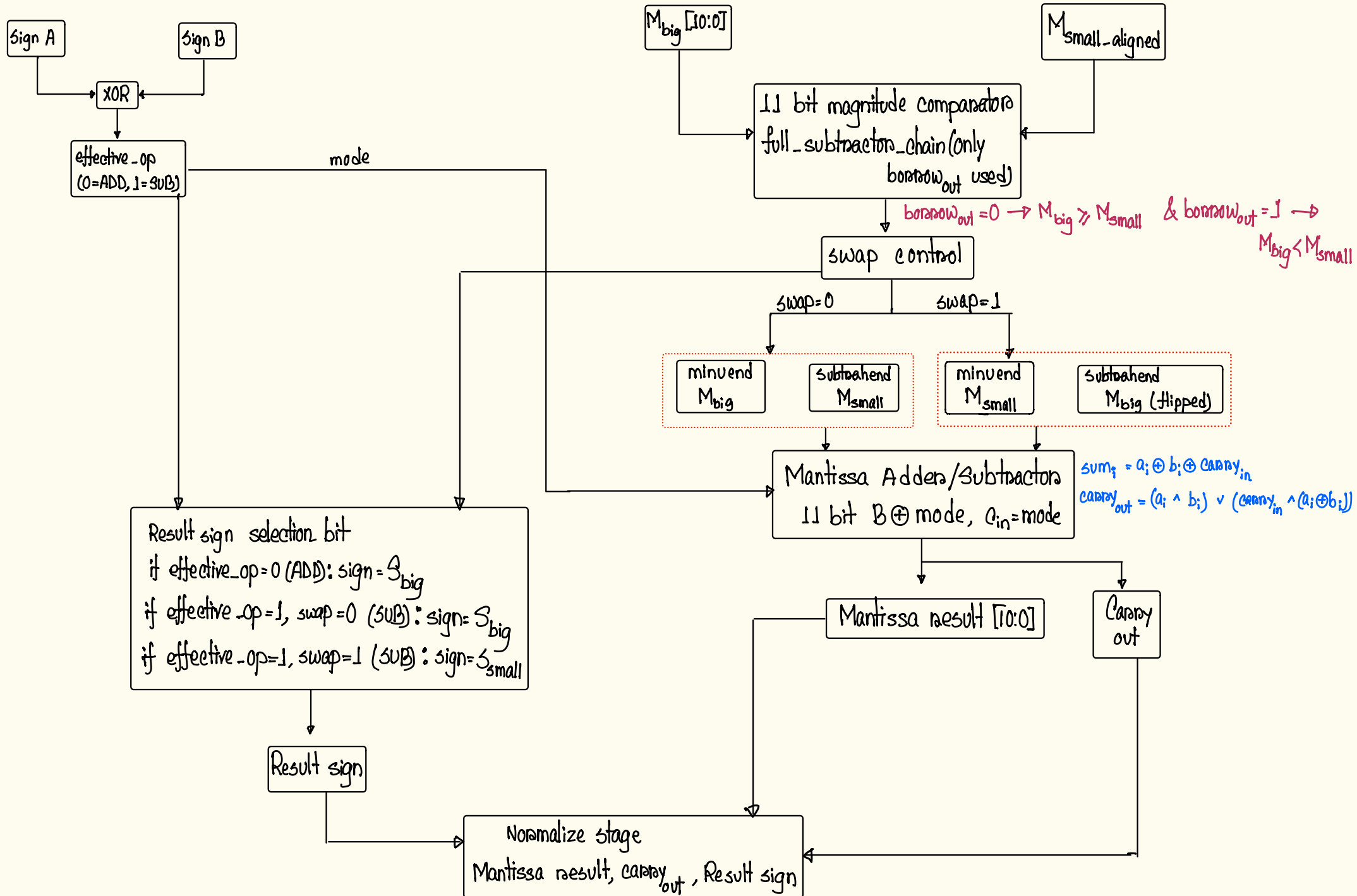
$$out = (in_0 \wedge \overline{sel}) \vee (in_1 \wedge sel)$$

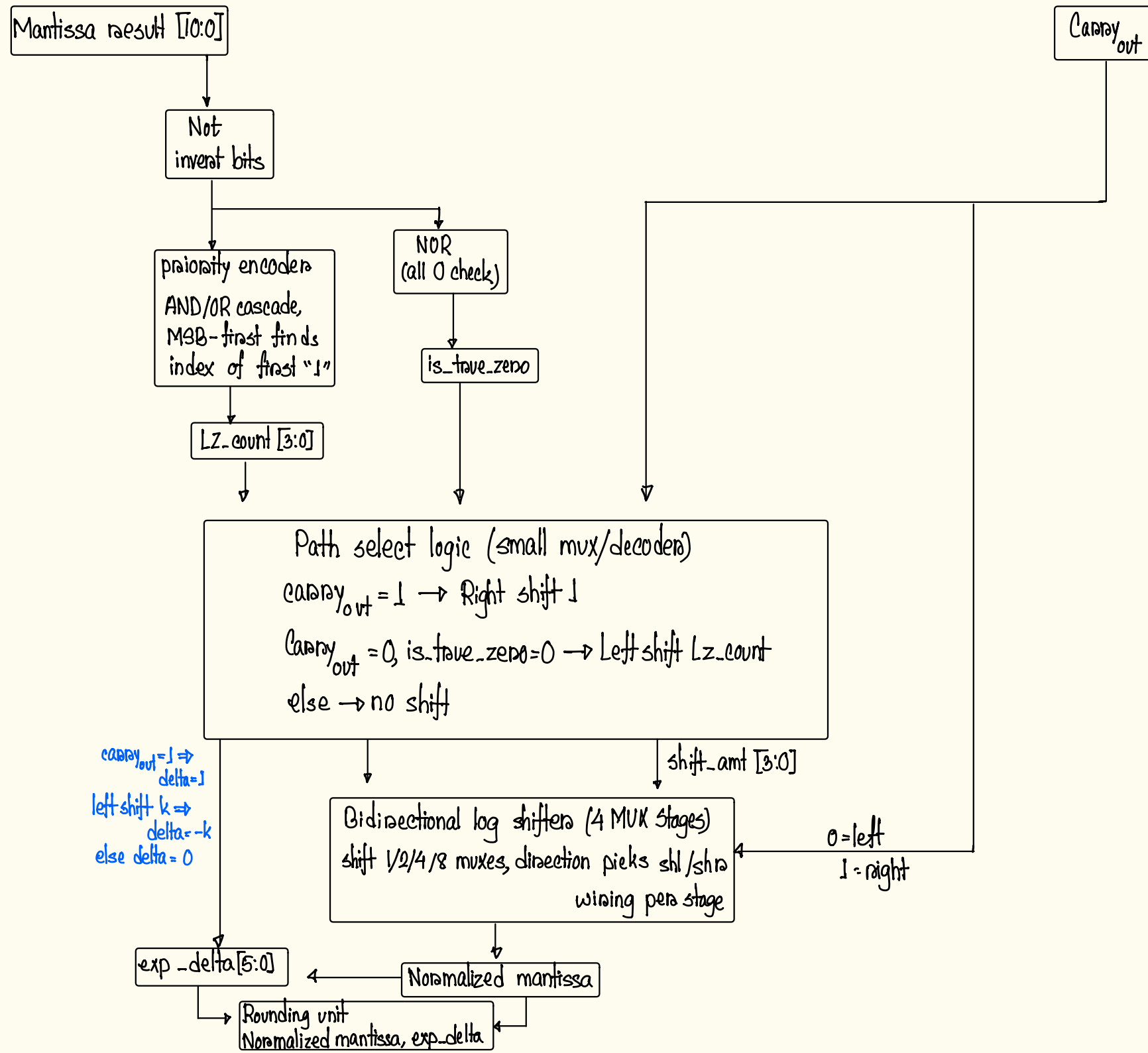
in_0 & in_1 are just connected to MUX's two inputs

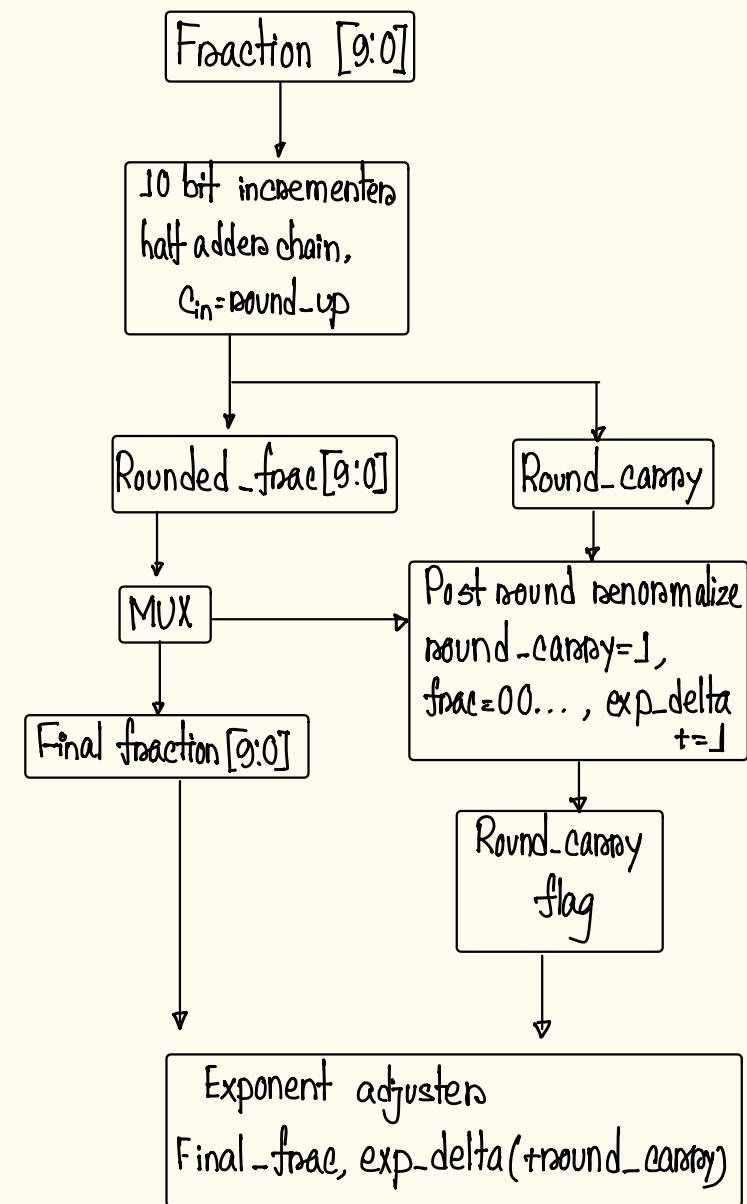
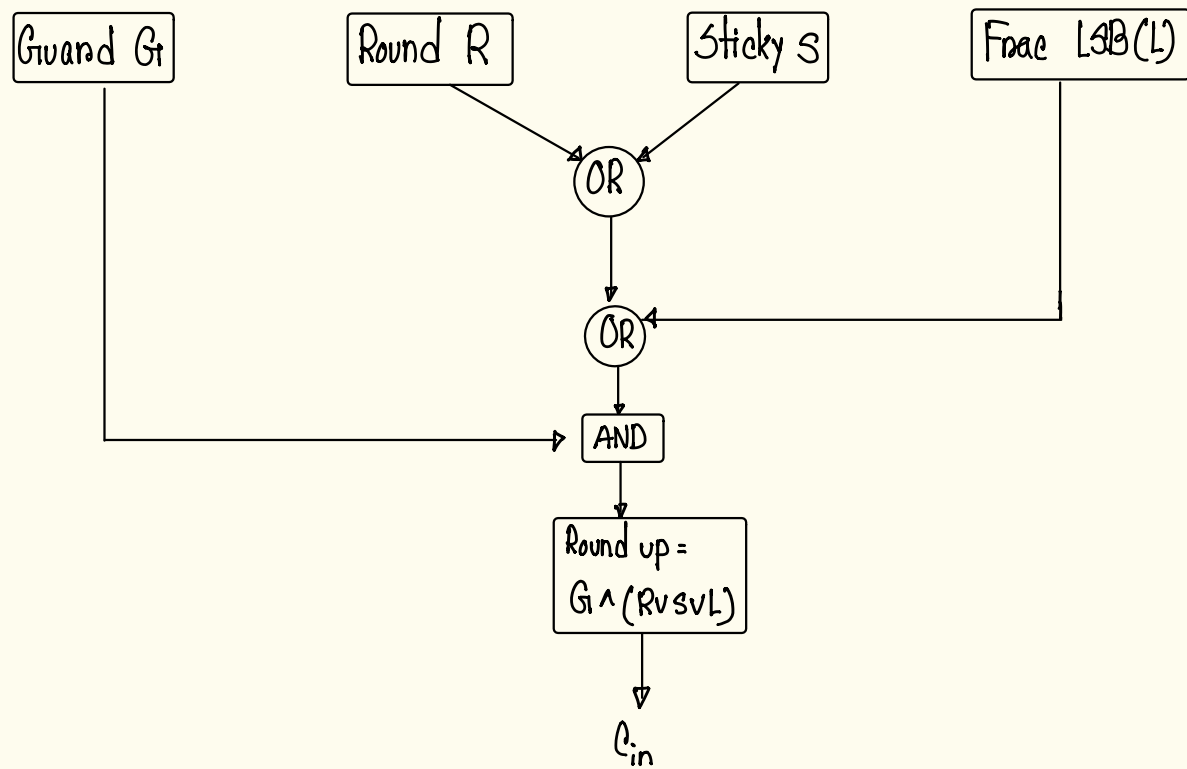
* $shamt-i \rightarrow$ shift by i

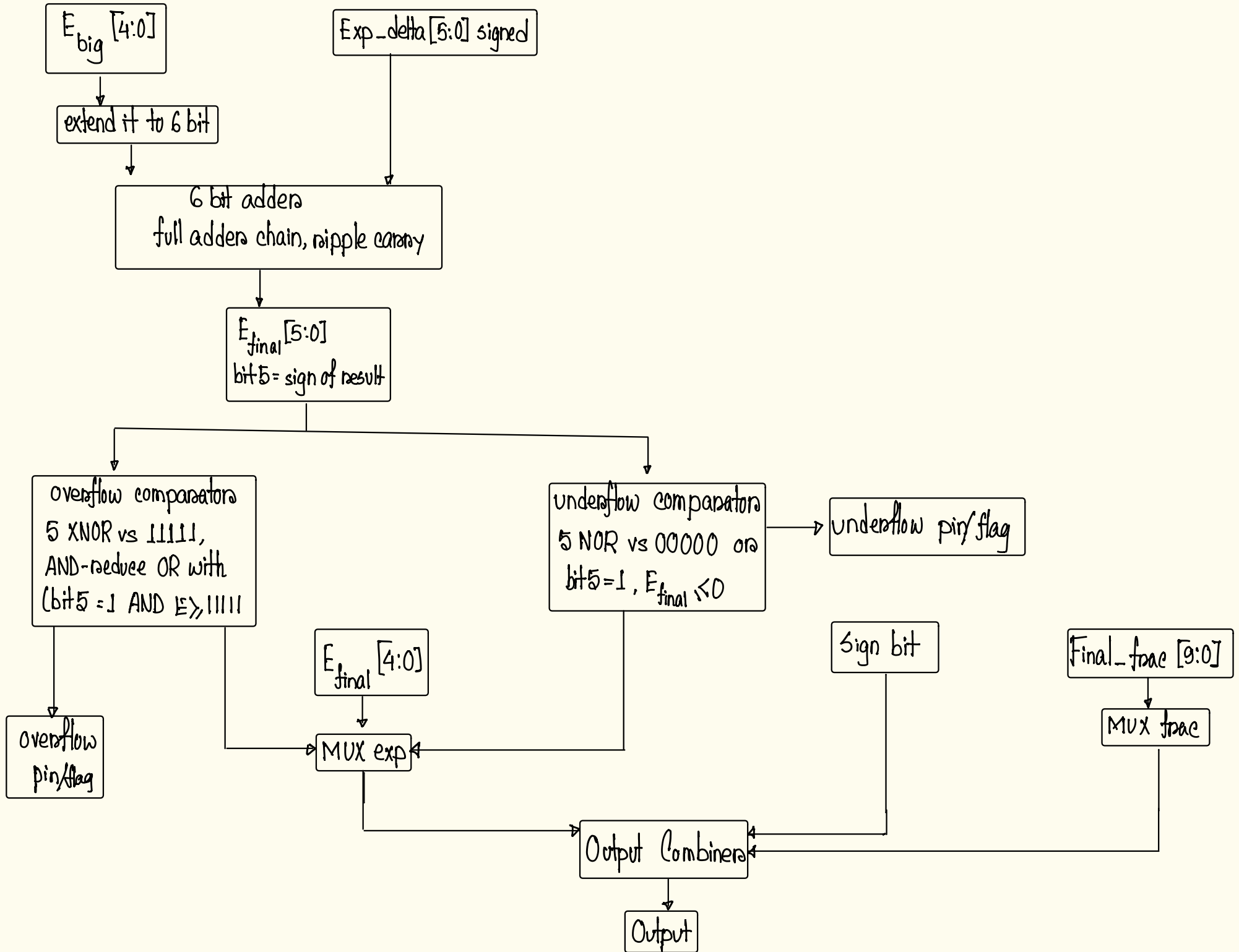
* $drop$ will only work if the MUX is shifted

* 11 bit is working mantissa,
the next bit down $\rightarrow$ guard bit
the bit below that is $\rightarrow$ round bit









$$10.0 + 0.029296875$$

$$A = 10.0$$

$$1010.0_2$$

$$= 1.010_2 \times 2^3 \rightarrow \text{exp (biased)} = 3 + 31 = \underline{34}$$

$$\begin{array}{c} 1 \ 6 \ 9 \\ \hline \text{bias} = 2^{6-1} - 1 \\ = 32 - 1 = 31 \end{array}$$

$$100010$$

$$A = 0 \ 100010 \ 0100000000$$

$$B = 0.029296875$$

$$= 1.111 \times 2^{-6} = 1.111 \times 2^{(25)} \rightarrow \text{exp (biased)}$$

$$34 - 25 = 9$$

$$B \gg 9$$

$$1 \rightarrow \text{field}$$

$$1 \rightarrow G$$

$$1 \rightarrow R$$

$$\pm \rightarrow S$$

$$\text{Fraction} =$$

$$000000001$$

$$1.0100000000$$

$$0.0000000001$$

$$1.0100000001$$

$$+1$$

$$\text{exp} = 100010$$

$$34 - 31$$

$$= 3$$

$$\text{Fraction}$$

$$1.0100000010 \rightarrow 1 + 2^{-2} + 2^{-8} = 1.25390625 \times 2^3 = 10.03125$$

$$\text{Add/sub-lib} \rightarrow \text{add/sub?}$$

$$\text{add/sub} \rightarrow 1 \text{ bit}$$

$$\text{but Normalizer} \rightarrow 16 \text{ bit}$$

$$40+12 = ?$$

1-5-6 format

$$40.0_{10} = (101000)_2 = 1.010000_2 \times 2^5 \quad \text{bias} = 2^{5-1} - 1 = 15 \quad \text{--- (1)}$$

$$12.0_{10} = (1100)_2 = 1.100000_2 \times 2^3 \quad D = 5 - 3 = 2$$

$$= 0.011000 \times 2^5 \quad \text{--- (2)}$$

(1) + (2) $\Rightarrow$

$$\begin{array}{r} 1.010000 \\ 0.011000 \\ \hline 1.101000 \end{array} \times 2^5 = \overset{32}{1} \overset{16}{1} \overset{4}{0} 100 = 52$$

$$14+2 = ?$$

1-4-8 format

$$14.0_{10} = (1110)_2 = 1.11000000_2 \times 2^3 \quad \text{--- (1)} \quad \text{bias} = 2^3 - 1 = 7$$

$$2.0_{10} = (10)_2 = 1.00000000_2 \times 2^1 \gg 2 \quad D = 3 - 1 = 2$$

$$= 0.01000000_2 \times 2^3 \quad \text{--- (2)}$$

$$(1) + (2) \Rightarrow 10.00000000_2 \times 2^3 = 1.00000000_2 \times 2^4 = 10000_2 = 16_{10}$$

$$64+3=9$$

1+6+4 format

$$64_{10} = 1000000_2 = 1.00000 \times 2^6$$

$$\text{bias} = 2^5 - 1 = 31$$

$$3_{10} = 11_2 = 1.1000 \times 2^1 \gg 5$$

$$D = 6 - 1 = 5$$

$G \wedge (R \vee S)$

$$= 0.0000 \underline{110} \rightarrow S$$

$$\text{Round up} = 1 \wedge (1 \vee 0) = 1$$

$$1.00000000$$

$$0.0000110$$

$$\text{exp} = 6 + 31 = 37 = 100101$$

$$\underline{1.0000110} \rightarrow \text{fraction} = 0001 \text{ (Round up)} = 1000100$$

$$0 \ 100101 \ 0001$$

$$= 68 \text{ (Ans)}$$

$$4.0078125 + 0.00390625$$

1+3+9 format

$$4_{10} = 100_2$$

$$4.0078125 = 1.000000001_2 \times (2^2)^{-0} 2^5 \rightarrow \textcircled{1}$$

$$0.0078125 = 0.00000001$$

$$0.00390625 = 0.0000000010_2 = 1.0000000000 \times (2^{-5})^{-0} 2^{-8} \rightarrow \textcircled{2^{-5}}$$

$$\text{exp} = 2^2 - 1 = 3$$

$$D = 5 - (-5) = 10$$

$$0.00390625_{10} \gg 10 = 0.00000000000100 \xrightarrow{\text{G} \rightarrow \text{R} \rightarrow \text{S}} \times 2^5 \quad \text{--- (2)}$$

① + ② $\Rightarrow$

$$1.0000000001100$$

$$1.000000010_2 \times 2^{(5)} \xrightarrow{2} = 1.000001000_2$$

$$= 4.015625$$

$$0 \quad 101 \quad 000000010$$

$$16 + 0.015625 = 9$$

1+5+9 format

$$16_{10} = 10000_2 = 1.000000000_2 \times 2^{(4)} \rightarrow 19 \quad \text{bias} = 2^4 - 1 = 15$$

$$0.015625_{10} = 0.000001000_2 = 1.000000000_2 \times 2^{(-6)} \rightarrow 9$$

$\gg 10$

$$D = 19 - 9 = 10$$

$$= 0.000000000100_2 \times 2^{19}$$

$$G = 1, R = 0, S = 0$$

Round up = of Big
 $G \wedge (R \vee S \vee \text{LSB})$
 $= 1 \wedge (0 \vee 0 \vee 1)$
 $= 1$

$$1.0000000000100_2 \times 2^{19}$$

$$LSB = 0$$

$$Round = G \wedge (R \vee S \vee L)$$

$$\Rightarrow 1.0000000000_2 \times 2^{(19) \rightarrow 4} = 16$$

$$= 0$$

$$Exp = 10011$$

$$0 \quad 10011 \quad 0000000000$$

$$frac = 0000000000$$

$$0.15625 + (-0.125) = ?$$

$$1 + 8 + 2$$

$$0.15625_{10} = 0.00101_2$$

$$bias = 2^7 - 1 = 127$$

$$= 1.01_2 \times 2^{-3} \quad sign = 0$$

$$D = 0$$

$$-0.125_{10} = 1.00 \times 2^{-3} \quad sign = 1$$

$$\begin{array}{r} 1.01 \\ 1.00 \\ \hline \end{array}$$

$$0.01 \times 2^{-3}$$

$$-5 + 127 = 122$$

$$0 \quad 01111010 \quad 00$$

$$\Rightarrow 1.00_2 \times 2^{(5) \rightarrow 4}$$

$$G = 0, R = 0, S = 0$$

$$\Rightarrow 0.03125_{10}$$

No Round

$$7.96875 + 0.0234375 = ?$$

$$1+4+7$$

$$7_{10} = 111_2$$

$$7.96875_{10} = 1111111_2 \times 2^{\textcircled{2} \rightarrow 9} \quad \text{bias} = 2^3 - 1 = 7$$

$$0.96875_{10} = 0.1111100_2$$

$$0.0234375_{10} = 0.0000011_2 = 1.1000000_2 \times 2^{\textcircled{-6} \rightarrow 1} \gg 8$$

$$D = 9 - 1 = 8$$

$$0.0000000110 \times 2^9 \quad \text{---} \textcircled{2}$$

$$\textcircled{1} + \textcircled{2} \Rightarrow$$

$$1.111111110 \times 2^{\textcircled{9} \rightarrow 2}$$

$$\hookrightarrow 10.0000000_2 \times 2^2$$

$$= 1.0000000_2 \times 2^{\textcircled{5} \rightarrow 10} = 8_{10}$$

$$LSD = 1, G = 1, R = 1, S = 0$$

$$\text{Round up} = G \wedge (R \vee S \vee L) = 1$$

$$0 \quad 1010$$

$$00000000$$

$$16_{10} + 0.0625_{10} = 9_0$$

$$16_{10} = 10000_2 = 1.0000 \times 2^{(4)} \rightarrow 35 \quad - (1)$$

$$0.0625_{10} = 0.0001_2 = 1.0000 \times 2^{(4)} \rightarrow 27 \gg 8$$

$$= 0.000000001000 \times 2^{35} - (2)$$

G R S

$$(1) + (2) \Rightarrow 1.0000 \times 2^{35} = 1.0000 \times 2^4 = 16$$

$$0 \quad 100011 \quad 000$$

$$-8 + (-2) = 2$$

$$-8_{10} = 100_2 = 1.000000 \times 2^{(3)} \rightarrow 18 \quad \text{sign} = 1 \quad - (1)$$

$$-2_{10} = 10_2 = 1.000000 \times 2^{(1)} \rightarrow 16 \gg 2 \quad \text{sign} = 1$$

$$= 0.010000 \times 2^{18} - (2)$$

$$1+6+3$$

$$\text{bias} = 2^5 - 1 = 31$$

$$D = 35 - 27 = 8$$

$G=0 \rightarrow$ Round down/truncate

$$1+5+5$$

$$\text{bias} = 2^4 - 1 = 15$$

$$D = 18 - 16 = 2$$

① + ② $\Rightarrow$

$$1'01000 \times 2^{\textcircled{18} \rightarrow 3} = 1010_2 = -10$$

$$1 \ 010010 \quad 01000$$

$$7.875 + 0.09375 = ?$$

1+3+5 format

$$\text{bias} = 2^2 - 1 = 3$$

$$7.875_{10} = 111.11100_2 = 1.1111_2 \times 2^{\textcircled{2} \rightarrow 5} \rightarrow \textcircled{1}$$

$$0.09375_{10} = 0.00011_2 = 1.10000 \times 2^{\textcircled{4} \rightarrow -1} \gg 6$$

$$D = 5 + 1 = 6$$

$$= 0.000000\underline{110} \times 2^5 - \textcircled{2}$$

G R S

$$G=1, R=1, S=0$$

Round = 1

① + ② $\Rightarrow$

$$1.1111110 \times 2^5$$

$$1.000000 \times 2^{\textcircled{6} \rightarrow 3}$$

= 8

0	110	00000
---	-----	-------

$$5 + 1.5 = 9$$

$$5_{10} = 101_2 = 1.010000_2 \times 2^{\textcircled{2} \rightarrow 9} - \textcircled{1}$$

$$1.5 = 1.10000000_2 \times 2^{\textcircled{0} \rightarrow 7} \gg 2$$

$$= 0.011000 \times 2^9 - \textcircled{2}$$

$$\textcircled{1} + \textcircled{2} \Rightarrow 1.101000 \times 2^{\textcircled{0} \rightarrow 2} = 110.1_2 = 6.5$$

$$0 \quad 1001 \quad 101000 \quad (\text{bias})$$

$$0 \quad 0010 \quad 101000 \quad (\text{unbiased})$$

$$12 + 6 = 9$$

$$12_{10} = 1100_2 = 1.10000000_2 \times 2^3 - \textcircled{1}$$

$$6_{10} = 110_2 = 1.10000000_2 \times 2^1 \gg 1$$

$$= 0.11000000 \times 2^3 - \textcircled{2}$$

$$1 + 4 + 6$$

$$\text{bias} = 2^3 - 1 = 7$$

$$D = 9 - 7 = 2$$

$$1 + 5 + 7$$

$$\text{bias} = 2^4 - 1 = 15$$

$$D = 3 - 2 = 1$$

$$\begin{aligned}
 ① + ② &\Rightarrow 10.0100000_2 \times 2^3 \\
 &= 1.0010000_2 \times 2^4 \\
 &= 10010_2 = 18
 \end{aligned}$$

Problem 3. 8 + 0.0234375

Format: 1 sign + 4 exponent (two's complement, unbiased) + 8 fraction = 13 bits total.

$$8_{10} = 100_2 = 1.00000000_2 \times 2^{\textcircled{3} \rightarrow 9}$$

1+4+8

$$\text{bias} = 2^3 - 1 = 7$$

$$\begin{aligned}
 0.0234375_{10} &= 0.00000110_2 = 1.10000000_2 \times 2^{\textcircled{-6} \rightarrow 1} \\
 &= 0.000000001100_2 \times 2^9 \\
 &\quad \quad \quad \text{G R S}
 \end{aligned}$$

$$D = 9 - 1 = 8$$

$$\begin{aligned}
 \text{Round} &= 1 \wedge (0 \vee 0 \vee 1) \\
 &= 1
 \end{aligned}$$

$$\begin{aligned}
 &1.00000001100_2 \times 2^9 \\
 &= 1.00000010_2 \times 2^9
 \end{aligned}$$

0 1001 00000010

Problem 4. $0.53125 + 0.015625$

Format: 1 sign + 6 exponent (two's complement, unbiased) + 4 fraction = 11 bits total.

$$1+6+4$$

$$\text{bias} = 2^5 - 1 = 31$$

$$0.53125_{10} = 0.10001_2 = 1.0001_2 \times 2^{-1} \rightarrow 30 - ①$$

$$0.015625_{10} = 0.000001_2 = 1.0000_2 \times 2^{-6} \rightarrow 25 \gg 5$$

$$D = 30 - 25 = 5$$

$$① + ② \Rightarrow$$

$$1.0001100_2 \times 2^{30}$$

$$= 0.0000100 \times 2^{30} - ②$$

$\frac{G}{G} \frac{R}{R} \frac{S}{S}$

$$G=1, R=S=0, L=1$$

$$\text{Round} = 1$$

$$1.0010_2 \times 2^{30}$$

$$0011100010$$

Problem 5. $0.0859375 + (-0.0625)$

Format: 1 sign + 5 exponent (two's complement, unbiased) + 3 fraction = 9 bits total.

$$1+5+3$$

$$\text{bias} = 2^4 - 1 = 15$$

$$0.0859375_{10} = 0.0001011_2 = 1.011_2 \times 2^{-4} \rightarrow 11 \text{ sign} = 0$$

$$D = 11 - 11 = 0$$

$$0.0625_{10} = 0.0001_2 = 1.000_2 \times 2^{-4} \rightarrow 11 \text{ sign} = 1$$

$$\begin{array}{r} 1.011 \\ - 1.000 \\ \hline \end{array}$$

$$0.011 \times 2^{11}$$

$$\approx 1.100 \times 2^9$$

$$0 \quad 01001 \quad 100$$